

Assignment for Data Scientist Applicant: Text Classification Using IMDb Review Dataset

Objective

The goal of this assignment is to assess your ability to handle text data and build classification models using different methods. You will use the IMDb Movie Reviews dataset for this task.

Dataset

- **IMDb Movie Reviews:** This dataset contains 100,000 movie reviews labeled as positive or negative.

Steps

1. Data Loading

- Load the IMDb Movie Reviews dataset.
- Split the dataset into training and testing sets.

2. Data Preprocessing

- Perform text cleaning (e.g., removing HTML tags, special characters, and converting text to lowercase).
- Tokenize the text data.
- Handle stop words by removing common non-informative words.
- Implement lemmatization or stemming to reduce words to their root forms.

3. Feature Extraction

- **Bag of Words (BoW):**
 - Convert text data into numerical features using the Bag of Words method.
- **TF-IDF (Term Frequency-Inverse Document Frequency):**
 - Convert text data into numerical features using the TF-IDF method.
- **Word Embeddings:**
 - Use pre-trained word embeddings (e.g., Word2Vec, GloVe) to convert text data into numerical features.
- **Deep Learning Embeddings:**
 - Use advanced deep learning models like BERT for feature extraction.

4. Model Training

- Train at least three different classification models:
 - **Naive Bayes**
 - **Support Vector Machine (SVM)**

- **Logistic Regression**
 - Optionally, try other models like Random Forest, Gradient Boosting, or deep learning models such as LSTM or CNN.
- For each model, handle class imbalance by:
 - Using resampling techniques (e.g., SMOTE).
 - Adjusting class weights.
- 5. **Model Evaluation**
 - Evaluate the performance of each model using appropriate metrics:
 - Accuracy
 - Precision
 - Recall
 - F1-Score
 - AUC-ROC
 - Visualize the confusion matrix for each model.
 - Discuss the pros and cons of each method used.
- 6. **Hyperparameter Tuning**
 - Perform hyperparameter tuning using methods such as Grid Search or Random Search to improve model performance.
- 7. **Documentation**
 - Document the entire process, including the rationale behind each preprocessing step, feature extraction method, and model choice.
 - Clearly explain how you handled class imbalance and the impact of the techniques used.
 - Include any challenges faced and how you addressed them.
 - Provide well-commented code for reproducibility.

Deliverables

1. **Code:** A well-organized Jupyter notebook or Python script containing all the code used for data loading, preprocessing, feature extraction, model training, and evaluation.
2. **Report:** A detailed report explaining your approach, findings, and conclusions. This should include plots, tables, and any other visualizations that support your analysis.
3. **Model Files:** Save and submit the trained model files if applicable.

Evaluation Criteria

- **Correctness:** The code should run without errors and produce the expected results.
- **Completeness:** All steps of the assignment should be completed thoroughly.
- **Clarity:** The code should be well-commented and the report should clearly explain each step and the reasoning behind your choices.
- **Performance:** The effectiveness of the models and techniques used will be evaluated based on the provided metrics.
- **Innovation:** Any additional techniques or models that demonstrate advanced knowledge will be considered positively.

Good luck! We look forward to reviewing your submission.